

Dual-Representation Mondrian Conformal Assignment of Pressure-Transient GeoTypes: An Independent Reproduction and Extension for Fractured-Reservoir Flow

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Abstract—Fractured reservoirs exhibit recurring fluid-flow behaviours hidden in the shape of their pressure-transient responses. A recent study clusters geologically consistent discrete-fracture-network (DFN) ensembles into flow-behaviour classes and attributes each class to fracture-network properties. We first independently reproduce that pipeline, DFN pressure transients, Bourdet-derivative features, dynamic-time-warping (DTW) k -medoids clustering into a GeoType catalogue, and random-forest / TreeSHAP attribution, from permissively licensed components on the original open 4TU corpus and on analytic dual-porosity families. The reproduction recovers clean clusters, with the real transients clustering more cleanly (silhouette up to 0.86) than the analytic families (0.15–0.27), and it recovers the finding that fracture intensity ($\log I$) is a leading attributed control (top on the benchmark cases). We then extend the pipeline in two ways. First, we make the assignment *coverage-controlled*, which the base random-forest attribution is not, with a class-conditional (Mondrian) split-conformal assigner on a DTW nonconformity score that emits per-class p -values, a prediction set, and an out-of-catalogue flag (empirical coverage 0.80–0.94, straddling the 0.85 target, across the rich-method cases). Second, we apply a two-score (multi-view) *dual-representation Mondrian conformal* intersection that conformalizes jointly across two nonconformity scores, the DTW shape distance and a physics-descriptor implausibility from the attribution forest, and accepts a GeoType only in the per-class conjunction of the two prediction sets. This excludes curves that are shape-close to the wrong physics, which single-score conformal trades: across five rich-method cases the dual conjunction trades marginal coverage (e.g. $0.85 \rightarrow 0.73$, $0.94 \rightarrow 0.87$) for tighter, physics-consistent sets. The shape machinery is signal-agnostic and is exercised on real hydrogeology pumping-test field data as well as reservoir transients. The contribution is a faithful, independent reproduction plus a coverage-controlled assignment that fuses shape space and physics-descriptor space, with every number backed by a committed deterministic artifact.

Index Terms—fractured reservoirs, pressure-transient analysis, discrete fracture networks, dynamic time warping, k-medoids, conformal prediction, Mondrian, random forests, SHAP, reproducibility, geothermal, hydrogeology.

I. INTRODUCTION

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Preprint, 2026, version 2.1 (revised). Code, committed derived artifacts, and the workbench are released under MIT at https://github.com/fsantibanezleal/CAOS_RES_Pulso; the reusable shape machinery is the Apache-2.0 package `pygeotypes` (https://github.com/fsantibanezleal/CAOS_GeoTypes). No GPL workflow code or raw licensed data is redistributed.

THE pressure-transient response of a well is a fingerprint of the reservoir’s flow structure: its shape, and the shape of its Bourdet derivative [3], distinguishes homogeneous from dual-porosity behaviour and localizes the interporosity transition [2]. In fractured reservoirs the mapping from geology to transient is many-to-one and uncertain, so a natural question is whether the space of transients organizes into a small catalogue of recurring flow behaviours, a “geological DNA”, and if so, which fracture-network properties control membership. A recent study answers both by clustering geologically consistent discrete-fracture-network (DFN) ensembles into flow-behaviour classes and attributing them [1].

This paper reproduces that pipeline and extends it with a coverage-controlled assignment. Reproduction matters here for two reasons: the original workflow depends on a GPL-licensed code base we deliberately do not vendor, and a clustering-plus-attribution result is only trustworthy if an independent re-implementation on the same open data recovers it. We re-implement the shape machinery from permissively licensed components, DTW [5], k -medoids, silhouette selection [6], random forests [7], and TreeSHAP [8], as the open package `pygeotypes`, run it on the original 4TU corpus [13] and on analytic dual-porosity families, and confirm the clusters and the top attributed control. The extension addresses a gap in how such catalogues are *used*: random-forest attribution ranks controls but gives no guarantee on a new curve’s class, and a curve can be shape-close to the wrong physics. We therefore add conformal prediction [9], [10], first a class-conditional (Mondrian) split-conformal assigner on the shape distance, then a dual-representation variant that conformalizes jointly over shape and physics-descriptor scores. Our contributions are: (i) an independent, permissively licensed reproduction of the DFN-transient GeoType pipeline that recovers its clusters and fracture-intensity attribution; (ii) a coverage-controlled Mondrian conformal GeoType assigner; (iii) a dual-representation Mondrian conformal assigner that excludes right-shape-wrong-physics curves, with the coverage/tightness trade-off reported honestly.

II. REPRODUCED PIPELINE

A GeoType catalogue is built in four stages. *Transients and features*. Each DFN ensemble (or analytic study) yields pressure transients; from each we compute the log-time pressure

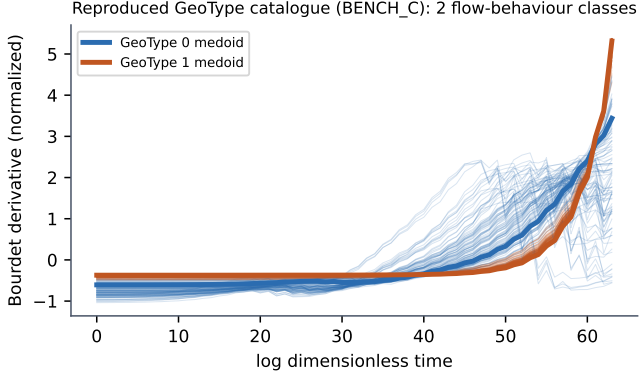


Fig. 1. A reproduced GeoType catalogue (case BENCH_C): the member Bourdet-derivative curves coloured by GeoType, with the two medoids in bold. Read from the committed `trace.json`.

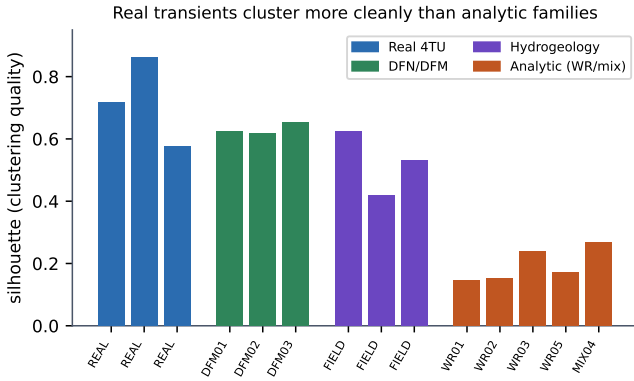


Fig. 2. Clustering quality (silhouette) per case, grouped by family. The real 4TU transients cluster more cleanly (up to 0.86) than the analytic Warren–Root families (0.15–0.27), with the DFN/DFM and hydrogeology cases in between. Read from the committed `trace.json` files.

and its Bourdet derivative [3], and for analytic dual-porosity references the Warren–Root response [2] is evaluated by Stehfest numerical Laplace inversion [4]. *Clustering*. Curves are compared by dynamic time warping [5] and grouped by k -medoids; the catalogue size K is chosen by a silhouette sweep [6], and each GeoType is represented by its medoid curve and a membership (Fig. 1). *Attribution*. A random forest [7] predicts the GeoType from physical descriptors, accuracy-gated (attribution is withheld when the forest cannot predict the labels), and TreeSHAP [8] ranks the controlling descriptors. *Assignment*. A new curve is classified against the baked medoids.

On the original open corpus the reproduction recovers the reported structure (Fig. 2): the real 4TU transients cluster more cleanly than the analytic families, and on the 4TU-benchmark cases fracture intensity ($\log I$, absorbing its $\rho > 0.9$ correlates) is the top attributed control (largest mean absolute SHAP), while on the real-4TU and analytic cases other descriptors (aperture, permeability, λ) lead, partially reproducing the paper’s fracture-intensity finding. On the analytic families the interporosity flow coefficient $\log_{10} \lambda$ dominates the assignment sensitivity, exactly as the physics predicts, since λ sets the tran-

sition timing that separates the dual-porosity behaviours. The same permissively licensed pipeline, exercised over twenty-one committed cases (analytic Warren–Root and mixture studies, real 4TU datasets, DFN/DFM network ensembles, and hydrogeology field sites; Appendix A), is a complete method ladder: classical pressure-transient analysis, DTW k -medoids clustering, silhouette-selected catalogue size, random-forest / TreeSHAP attribution, and the conformal assignment below.

III. COVERAGE-CONTROLLED ASSIGNMENT

A. Mondrian split-conformal on the shape score

Random-forest attribution ranks controls but offers no guarantee that a new curve is assigned to the right GeoType. We add a class-conditional (Mondrian) split-conformal assigner [9], [10]. The nonconformity score of a curve x for GeoType g is its DTW distance to the medoid, $s_{\text{shape}}(x, g)$. Calibrated per class on a disjoint slice, it yields a p -value $p_{\text{shape}}(x, g)$ and a prediction set $\{g : p_{\text{shape}}(x, g) > \alpha\}$; a curve matching no class is flagged out-of-catalogue. Across the rich-method cases the empirical marginal coverage is 0.80–0.94, straddling the target level, with an out-of-catalogue rate of 0.03–0.17: a coverage guarantee the base attribution cannot give.

B. Dual-representation Mondrian conformal assignment

A single score cannot express that a curve may be shape-close to the wrong physics: a transient that matches a medoid by DTW shape yet whose physical descriptors are atypical for that GeoType. We conformalize jointly across two representations. In addition to the shape score, define a descriptor score

$$s_{\text{desc}}(x, g) = 1 - \text{RF}(g \mid \text{descriptors}(x)), \quad (1)$$

the implausibility of x ’s physical descriptors under GeoType g from the attribution forest. Both are Mondrian-calibrated, giving p_{shape} and p_{desc} , and the dual prediction set is the per-class conjunction

$$\Gamma^\alpha(x) = \{g : p_{\text{shape}}(x, g) > \alpha \wedge p_{\text{desc}}(x, g) > \alpha\}. \quad (2)$$

A GeoType is accepted only if the curve is both shape-consistent and descriptor-consistent; a shape match with implausible physics is excluded. Intersecting independently calibrated conformal sets across representations is an established multi-view/multi-score construction [11], [12]; our contribution is its *application*, fusing a DTW shape score with a random-forest descriptor-implausibility score for pressure-transient GeoType assignment, and the honestly reported coverage/tightness trade-off. Because the per-class conjunction is a Bonferroni-style combination, its marginal coverage is only guaranteed at $1 - 2\alpha$ and in practice falls below the single-score $1 - \alpha$, which is precisely the coverage cost we report.

Fig. 3 reports the honest trade-off across five rich-method cases. The dual conjunction tightens the mean set size and flags right-shape-wrong-physics curves that shape-only conformal accepts, at the cost of some marginal coverage (for example $0.85 \rightarrow 0.73$ on BENCH_A and $0.94 \rightarrow 0.87$ on REAL_A). When descriptors are absent or degenerate (some

Dual conformal trades coverage for tighter, physics-consistent sets

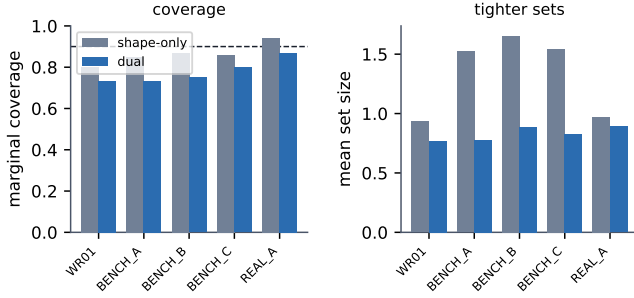


Fig. 3. Dual-representation vs shape-only Mondrian conformal across five rich-method cases. The dual conjunction trades marginal coverage (left) for tighter, physics-consistent prediction sets (right), excluding curves that are shape-close to the wrong physics. Read from the `attribution_plus.dual_conformal` block of each `trace.json`.

real cases), the dual layer degrades gracefully to shape-only and reports that it did.

IV. GENERALIZATION AND HONEST SCOPE

The shape machinery is signal-agnostic: any response signal whose shape reflects the underlying physical system, hydrogeology pumping tests, geothermal and thermal-response tests, can be catalogued and conformally assigned by the same code. We exercise it on real hydrogeology pumping-test field data (the Horkheim and Lauswiesen sites) alongside the reservoir transients, all through one pipeline (Appendix A).

Scope is reported plainly. This is a reproduction plus extension of a published pipeline, not a wholly new method for the base clustering and attribution, which are credited to the reproduced work. The DFN pressure-transient simulation on the full open-source flow engine is an offline dependency whose end-to-end integration is a later phase; the results here rest on the paper’s own committed corpus, analytic dual-porosity families, DFN ensembles, and field data, and the artifacts that lack a simulated transient carry an explicit pending flag rather than a fabricated curve. The dual-conformal coverage is a genuine trade: it buys physics-consistency and tighter sets at the price of marginal coverage, and both sides are shown. Every number is a committed, seed-deterministic artifact.

V. CONCLUSION

We independently reproduced, from permissively licensed components, a DFN-transient GeoType pipeline for fractured-reservoir flow, recovering its clusters (silhouette up to 0.86, real transients cleaner than analytic) and its fracture-intensity attribution, and extended it with a coverage-controlled Mondrian conformal assigner and, as the methodological contribution, a dual-representation Mondrian conformal assigner that fuses shape and physics-descriptor evidence to exclude right-shape-wrong-physics assignments. The result is a reproducible, coverage-controlled way to place a new pressure transient in a catalogue of flow behaviours and to say, with a guarantee, when it belongs to none.

DATA AND CODE AVAILABILITY

Code, the committed derived artifacts, and the workbench are at https://github.com/fsantibanezleal/CAOS_RES_Pulso (MIT); the reusable shape machinery is `pygeotypes` (Apache-2.0). The reference corpus is the open 4TU dataset of the reproduced study [13] (GPL-3.0, kept in a local vault, never committed); GPL workflow code is not vendored. Only compact derived artifacts are committed. The figures are regenerated deterministically from the committed traces by `manuscripts/dual-conformal-geotypes/figures/make_f`.

APPENDIX A PER-CASE RESULTS

The pipeline was run over twenty-one committed cases. Representative silhouette, out-of-catalogue rate, and (where the dual layer runs) the shape-only and dual coverage are: REAL_A 0.72 / 0.94 $\rightarrow$ 0.87 dual; REAL_B 0.86; REAL_C 0.58; BENCH_A 0.57 / 0.85 $\rightarrow$ 0.73; BENCH_B 0.48 / 0.87 $\rightarrow$ 0.75; BENCH_C 0.62 / 0.86 $\rightarrow$ 0.80; DFM01 0.62; DFM03 0.65; FIELD_horkheim 0.62; FIELD_laueswiesen 0.42; WR01 0.15 / 0.80 $\rightarrow$ 0.73; the analytic WR and mixture families 0.15–0.27. The full per-case table (silhouette, K , coverage, out-of-catalogue rate, mean set size, dual comparison, and attribution) is committed in each case’s `trace.json`.

REFERENCES

- [1] E. Kamel Targhi, G. Rongier, P.-O. Bruna, A. Daniilidis, and S. Geiger, “Unsupervised learning for geologically consistent fluid flow analysis in fractured reservoirs,” *Comput. Geosci.*, vol. 30, 57, 2026, doi:10.1007/s10596-026-10459-w.
- [2] J. E. Warren and P. J. Root, “The behavior of naturally fractured reservoirs,” *SPE J.*, vol. 3, no. 3, pp. 245–255, 1963, doi:10.2118/426-PA.
- [3] D. Bourdet, J. A. Ayoub, and Y. M. Pirard, “Use of pressure derivative in well-test interpretation,” *SPE Formation Eval.*, vol. 4, no. 2, pp. 293–302, 1989, doi:10.2118/12777-PA.
- [4] H. Stehfest, “Algorithm 368: Numerical inversion of Laplace transforms,” *Commun. ACM*, vol. 13, no. 1, pp. 47–49, 1970, doi:10.1145/361953.361969.
- [5] H. Sakoe and S. Chiba, “Dynamic programming algorithm optimization for spoken word recognition,” *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 26, no. 1, pp. 43–49, 1978, doi:10.1109/TASSP.1978.1163055.
- [6] P. J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987, doi:10.1016/0377-0427(87)90125-7.
- [7] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi:10.1023/A:1010933404324.
- [8] S. M. Lundberg *et al.*, “From local explanations to global understanding with explainable AI for trees,” *Nat. Mach. Intell.*, vol. 2, pp. 56–67, 2020, doi:10.1038/s42256-019-0138-9.
- [9] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005, doi:10.1007/b106715.
- [10] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023, doi:10.1561/22000000101.
- [11] E. Garcia-Ceja, “Multi-view conformal learning for heterogeneous sensor fusion,” 2024. arXiv:2402.12307.
- [12] C. Xu, Y. Yu, and H. Ren, “Aggregating conformal prediction sets via α -allocation,” 2025. arXiv:2511.12065.
- [13] E. Kamel Targhi *et al.*, “Data and code supporting: Unsupervised learning workflow for clustering pressure transient responses in fractured reservoirs (dataset),” 4TU.ResearchData, 2026, doi:10.4121/8291d285-025d-4724-988d-fc747a578c0a.